

Time Series Analysis

Conclusion

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Informatik



Outline

Mental model for time series projects.

Common mistakes & best practices.

Reading ML research papers.

Mental model for time series projects

Define the **learning task**: forecasting, classification, anomaly detection, clustering, representation learning.

Understand the data-generating process and temporal **structure of the data**.

- Sampling rate, seasonality, trends, regime changes.
- Multivariate vs univariate.
- Irregular sampling, missing values.

Choose **how time enters the model**: explicit features, sequence models (full-sequence vs window-based).

Design a time-consistent **evaluation strategy**.

- Match evaluation to the task e.g., horizon, detection delay, class imbalance.

Interpret results with **uncertainty** and **domain knowledge**.

- Check robustness and failure modes.
- Relate predictions to real-world processes.

Common mistakes – Data analysis

Ignoring **data quality** issues

- Missing values, outliers, and inconsistent timestamps.
- Data format mismatches (e.g., date-time parsing errors).

Insufficient exploratory analysis

- **No visualization**; missing key patterns, seasonality, or trends.
- Skipping basic summary statistics, leading to poor data understanding.

Data transformations: smoothing, differencing, normalization.

- Not considering **impact on data structure**: temporal leakage, semantic shift, distortion of patterns.

Lack of **domain knowledge**

- Over-reliance on AutoML e.g., automated feature extraction.
- Neglecting known external factors (e.g., holidays, policy changes, market shocks).

Best practices for data analysis

Remember the **garbage in, garbage out** principle.

Documentation: keep records of every data cleaning/transformation steps for reproducibility.

Perform **sanity checks** on the raw data, after pre-processing and after transformations.

Visualize more than less: use variety of plots (line charts, seasonal plots, ACF plots) to gain insights.

Contextualize the data

- Understand the data generation process.
- Interpret outliers, relate to real-world events.
- Ensure transformations and assumptions make sense.
- Interpret results with domain knowledge.

Common mistakes – Splitting

Messing the **temporal order**

- Using random splits as if samples were independent and identically distributed.
- Forecasting using future observations.

Splitting the dataset without considering **seasonal** cycles or **trend** patterns.

- A data split defines **what information the model is allowed to know**.
- Splitting is a modeling assumption about the future.

Selecting too short test period that doesn't capture **long-term patterns**.

Not considering **regime changes** in the data.

Data splits with **different distribution**.

Common mistakes – Temporal leakage

Target misalignment: input includes information from the same time step as the target (or from the future).

- Model ends up solving an easier, invalid problem.

Leakage sources:

- Random train/test splits on time-dependent data.
- Feature engineering using the full dataset.
- Scaling, normalization, or imputation fit on all data.
- Differencing or smoothing applied before splitting.
- Target misalignment in sliding windows.

At prediction time, the model must only use information that is **available in reality**.

Common mistakes – Evaluation workflow

Not reserving test data: performance evaluation on **data seen during training**.

Overfitting to the validation set by repeatedly adjusting parameters without a separate hold-out test set.

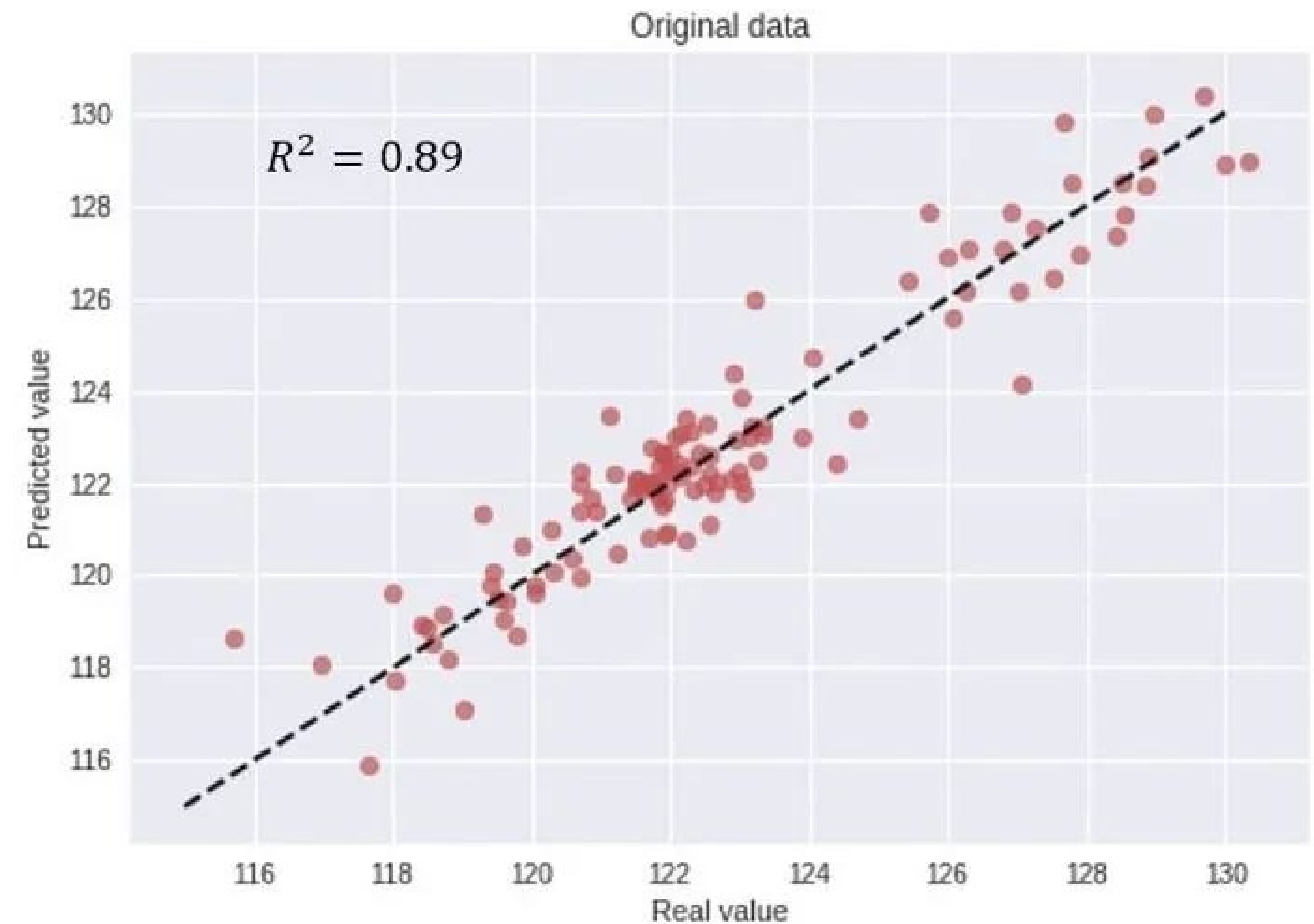
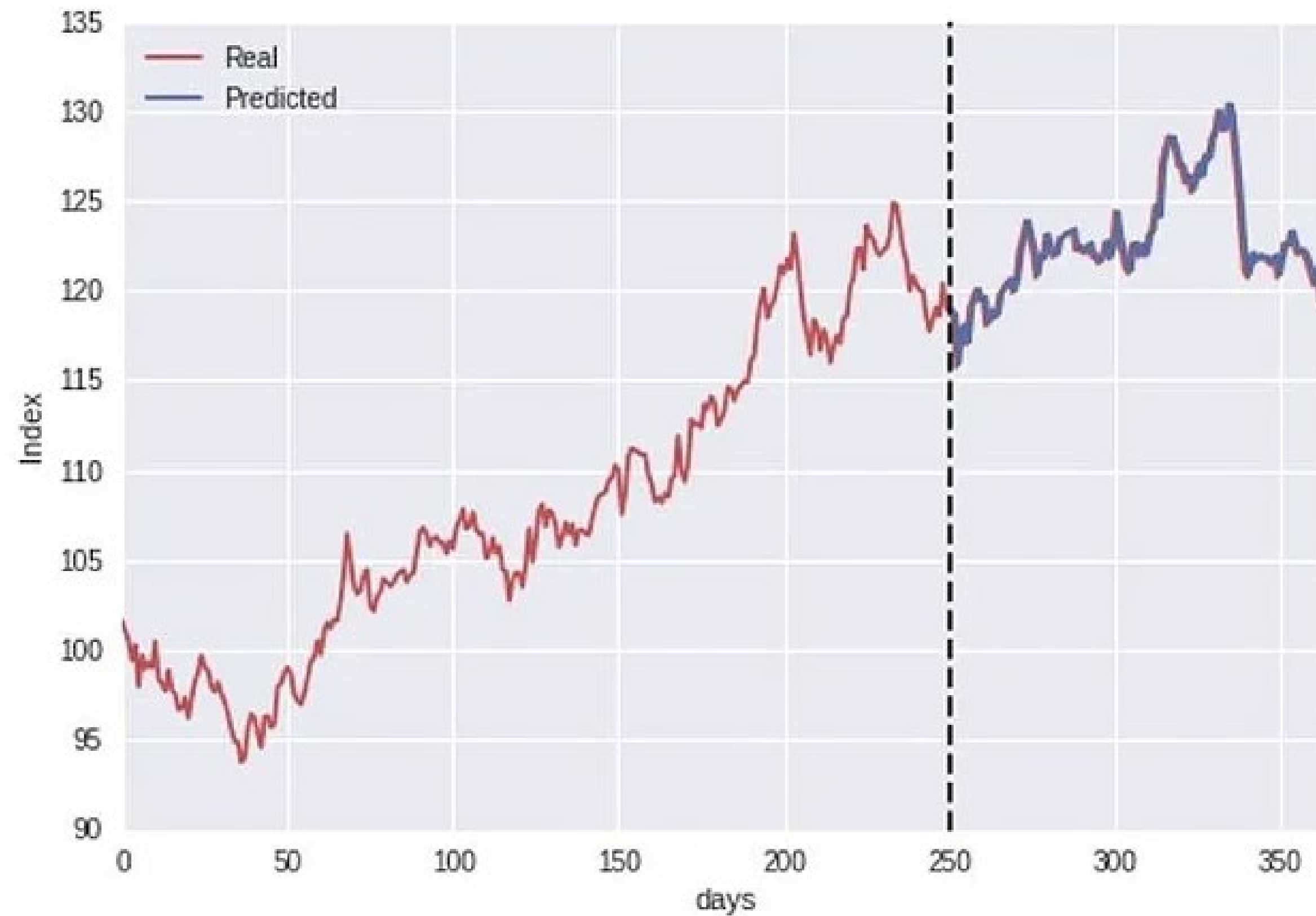
Not comparing against appropriate **baselines**, using inappropriate performance metrics.

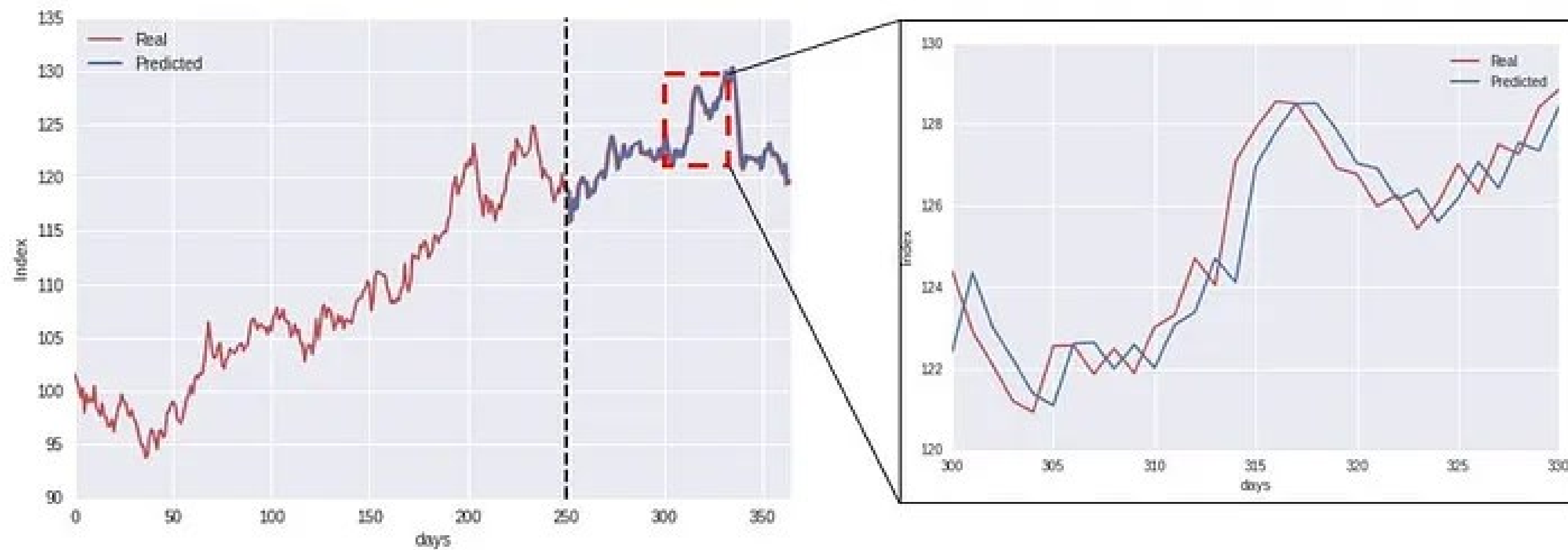
No estimation of **uncertainty**.

Using a single validation window instead of multiple rolling or expanding windows.

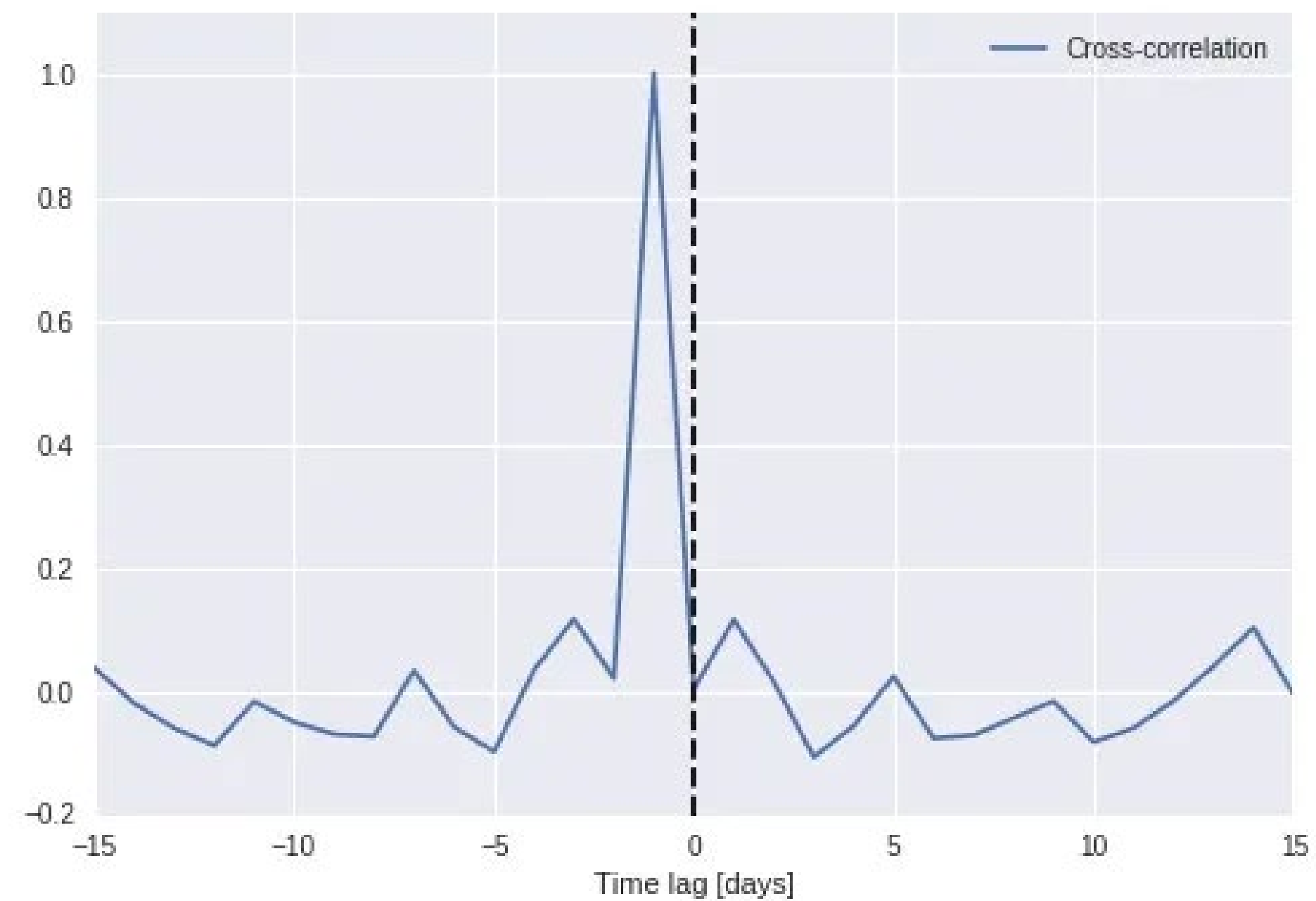
Not evaluating models on **out-of-distribution** samples e.g., data collected in different centers.

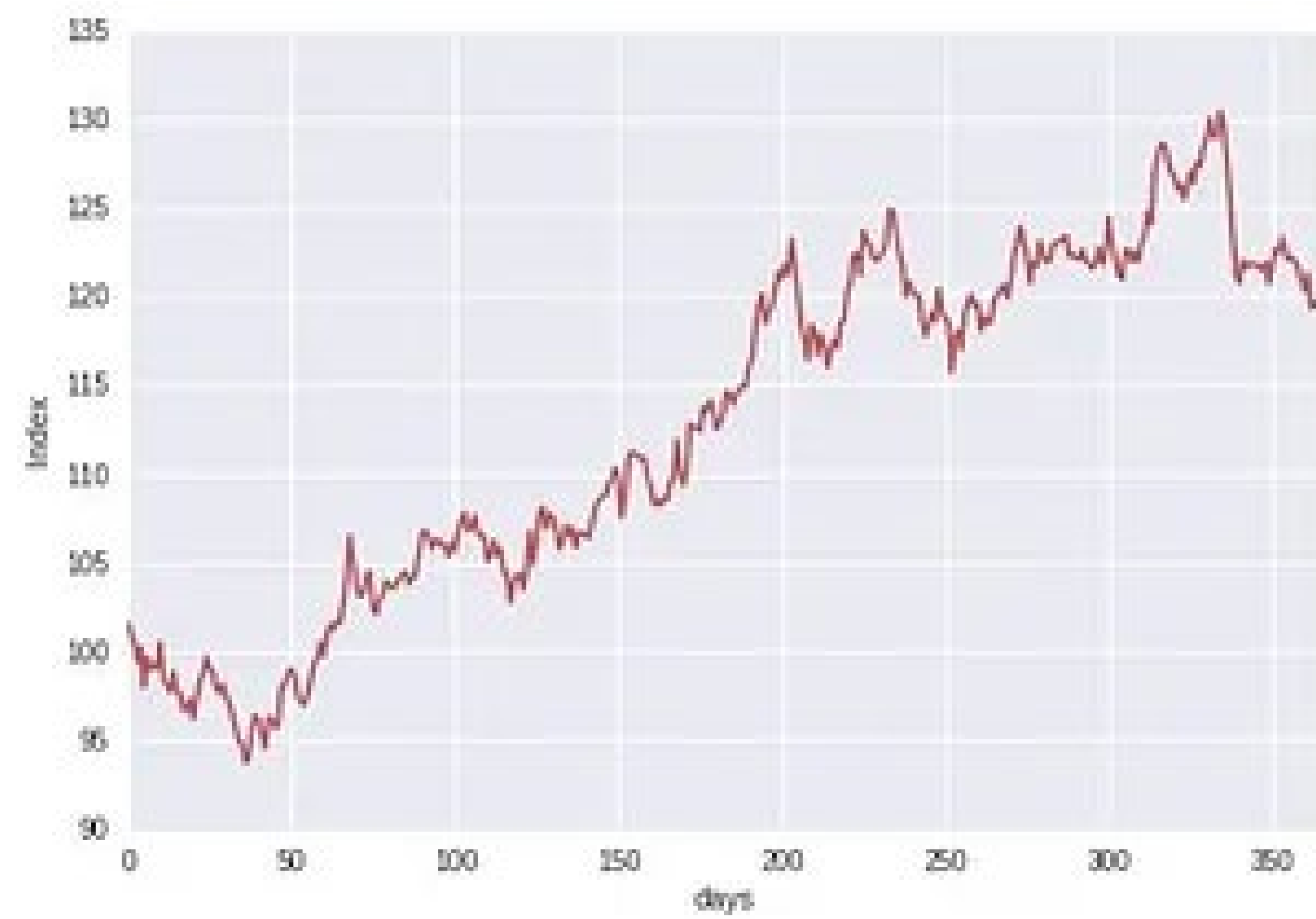
When performance are too good ...



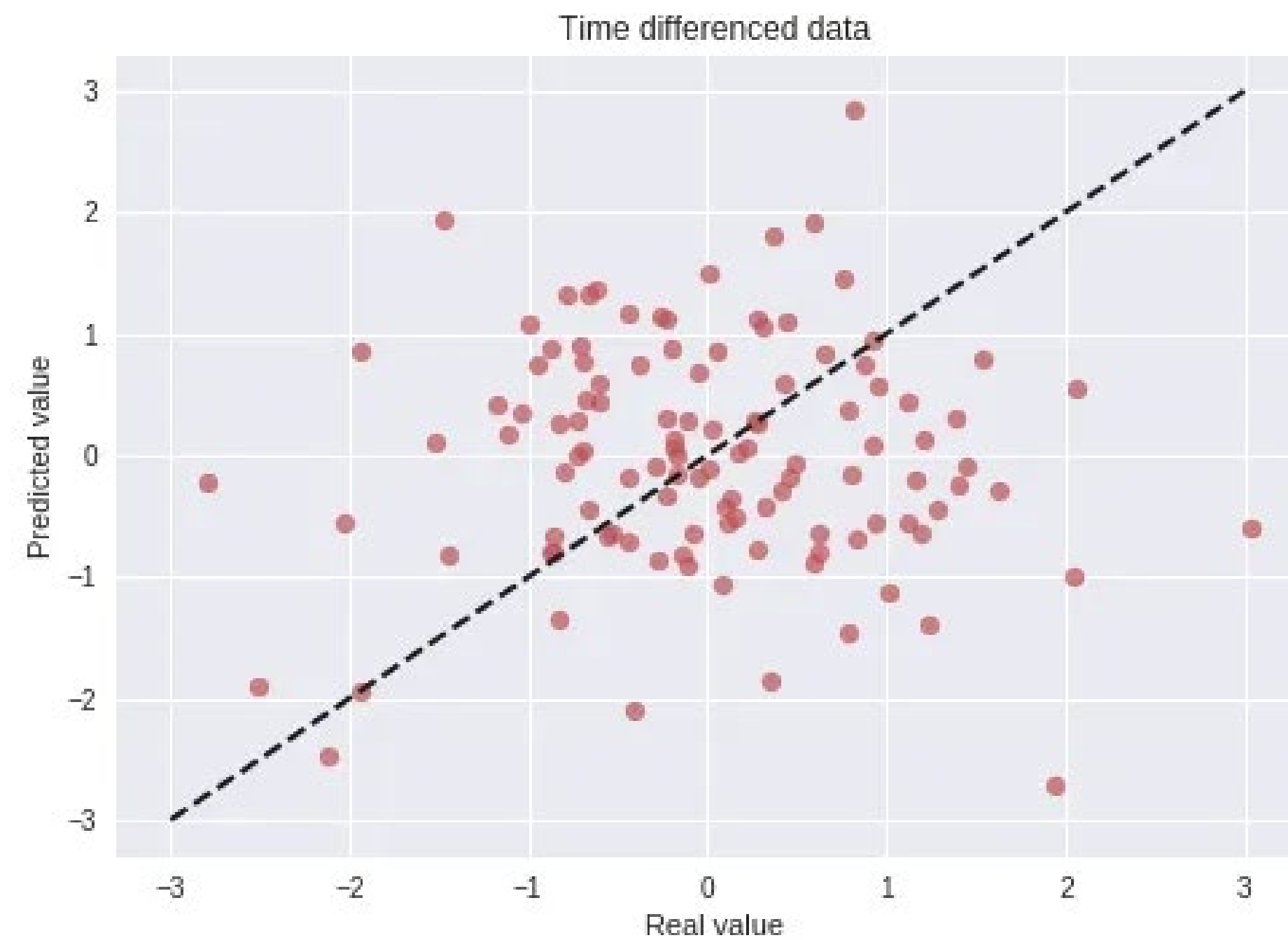
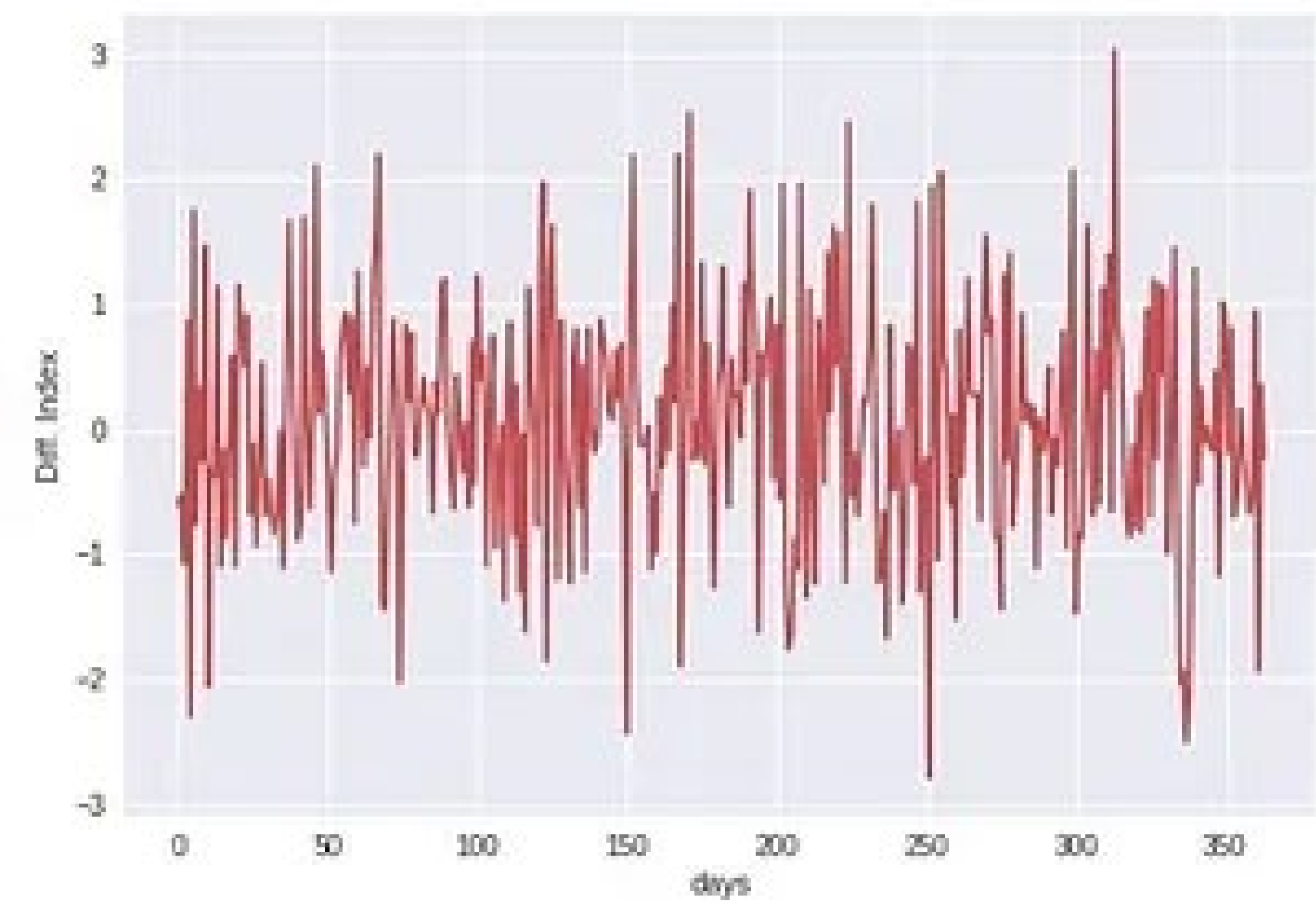


Cross-correlation between predicted and real value





Time differencing



When performance are too good ...

High performance is not a success signal, rather a **warning sign**.

Check **temporal alignment** of inputs and targets.

Inspect **cross-correlation** between the predicted and real series at non-zero lags.

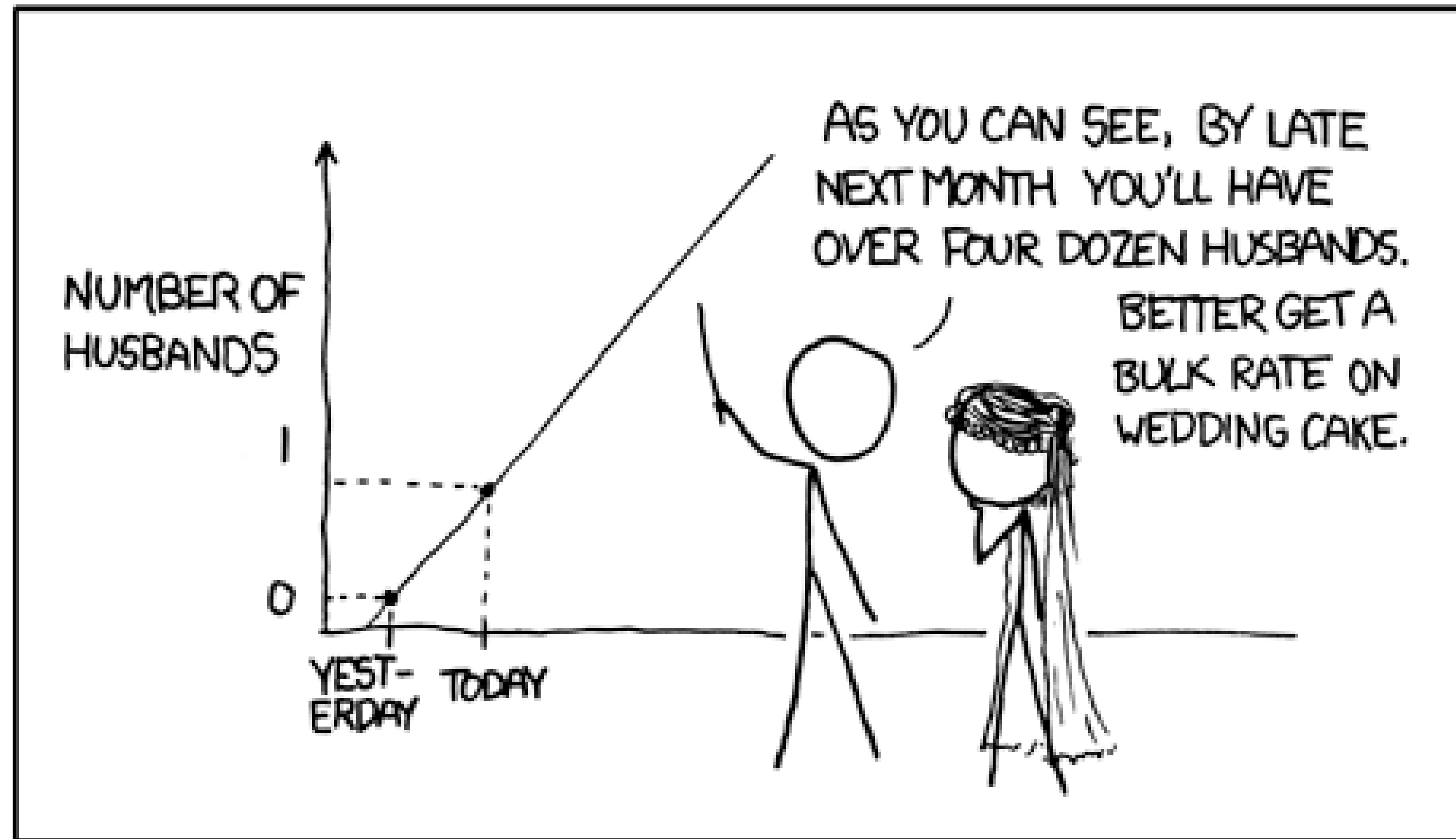
Compare against simple **baselines**: naïve, seasonal mean, random walk.

Verify that preprocessing was fit on **training data only**.

Ask: Could this result be obtained **without future information**?

Common mistake – Extrapolating linear regression

MY HOBBY: EXTRAPOLATING



Randall Munroe, <https://xkcd.com/605/>

Extrapolation is a modeling assumption

Linear trends are **local approximations**. Time is **unbounded**, data is not.

Linear regression assumes:

- Stable generative process.
- No regime change.
- **Short extrapolation horizon.**

Safe practice:

- Treat trend models as local.
- Validate on rolling future windows.
- Expect trend failure, plan for it.

Do not ask “will this extrapolate?” but “how quickly will this extrapolation break?”.

Common mistakes

Aliasing due to inappropriate sampling rate.

Granularity of timestamps: too coarse/fine can obscure patterns.

Misalignment between **forecast horizon** and actual needs.

Model **assumptions** and uncertainty evaluation: stationarity, error distribution etc.

Spurious correlations: mistaking random fluctuations for patterns.

Data spanning too short period of time may hide underlying patterns.

Failure to detect/model **regime changes**.

Choosing the right modeling paradigm

Classical models

- Explicit assumptions, probabilistic by design.
- Strong when data is scarce or interpretable uncertainty is required.

Machine learning:

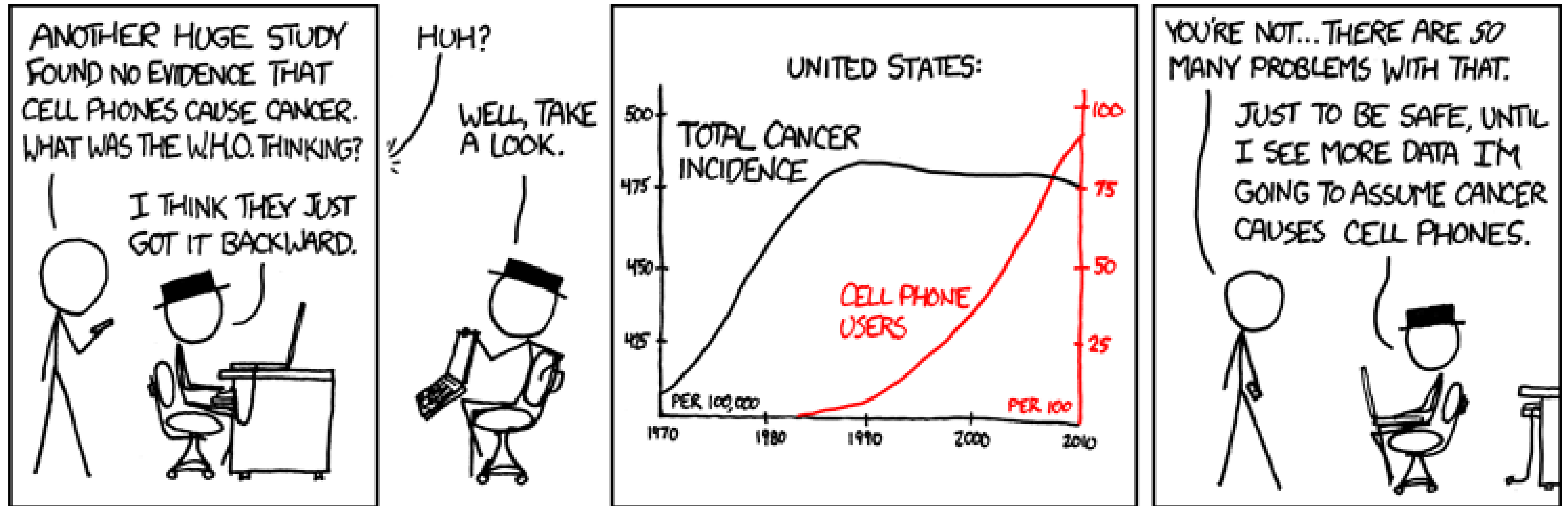
- Feature-driven, flexible, often the strongest baseline.
- Requires careful feature engineering to avoid leakage

Deep learning:

- Learns representations and long-range dependencies.
- Data-hungry, harder to debug, easier to overfit temporally.

Prefer the **simplest** model that satisfies horizon, uncertainty, and robustness **requirements**.

Correlation/causation fallacy



Randall Munroe, <https://xkcd.com/925/>

Reading ML research papers

A ML research paper is a structured document where researchers present

- A new method or algorithm
- Results comparing methods on real-world data
- Analysis of why their approach works (or doesn't)

Typical structure

- Abstract: short summary of the main idea and results.
- Introduction: the problem, why it matters, and what the paper contributes.
- Related work: previous approaches and how this paper is different.
- Method/approach: the technique, model architecture, or algorithm.
- Experiments & results: how well the method works, often using real data.
- Discussion: interpret the results, talks about limitations.
- Conclusion & future work: summarizes findings and suggests what's next.

Reading ML research papers – Why?

Reading research papers is **key** to become a good ML engineer.

Ensuring **skillset** remains relevant by staying **up-to-date** with latest trends and methods.

Better **understanding** of methodology assumptions, use-cases, strengths and limitations.

Inspiration for **real-world solutions**.

World economic forum: skills development and **lifelong learning** improve the employability of workers.

Finding ML research papers

Determine the **topics of interest** e.g., time series forecasting, pose estimation, summarization.

Finding research papers

- PapersWithCode, arxiv-sanity, Google Scholar, etc.
- Leading conferences/journals e.g. NeurIPS, ICML, ICLR, JMLR.

Leverage **reviews and surveys**

- **Keywords, nomenclature** and **taxonomy**.
- Extensive reference list.

Ensure a paper is **relevant**: check number of citations, review abstracts.

Explore **references** of relevant papers.

Reading ML research papers – How?

1. Understand the context and gain intuition on the paper: **title, abstract, and conclusion.**
 - Note keywords, datasets, methods, etc. (use tools like Logseq, Notion)
 - Gain overview of the paper's scope and objectives.
2. Familiarize yourself with the paper content: **introduction and figures.**
 - Presentation of the problem, past works, research gaps and paper contribution.
 - Develop your first intuition on the paper.
 - Studying figures at this stage enables a more critical perspective.
3. Deep reading: **full paper** from abstract to conclusion but **skipping complex math or formulations.**
 - Note key insights and takeaways along with skipped parts.
4. Final pass: **come back** to skipped notions/parts, use external sources to facilitate understanding.

Reading time series research critically

Check how:

- Data is split: are causality and task constraints respected?
- Baselines are chosen: are they appropriate, competitive, and fairly tuned?
- Horizons are evaluated: does it match the claimed scope?

Beware of:

- Results based on a single dataset or regime.
- Missing or weak baselines.
- Unclear preprocessing, feature engineering.

Prefer papers with:

- Ablation studies isolating the effect of each design choice.
- Evaluation across multiple datasets, horizons, or domains.
- Protocols that reflect realistic deployment conditions.